



## AQ3.2: Activity Questions 2 - Not Graded



This assignment will not be graded and is only for practice.

### Level 1

If  $(1, 2, 3) + v = (3, 2, 1)$  in the vector space  $R^3$  with usual addition and scalar multiplication, where  $v = (a, b, c)$ , then find the value of  $a + b + c$ .

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 0

1 point

Which of the following statements are related to the cancellation law for vectors  $x, y, z$  in a vector space.

Statement P:  $x + y = z + y \implies x = z$ .

Statement Q:  $x + y = z + y \implies x = y$ .

Statement R:  $x + y = z + y \implies y = z$ .

Statement S:  $x + y + z = x \implies y + z = 0$ .

Find the number of correct statements

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 2

1 point

1 point

Which of the following options is/are true for a non zero vector space  $V$ ?

☐

$x + y = z + y \implies x = z \forall x, y, z \in V$ .

☐

$ax = bx, \forall x \in V \implies a = b$ , where  $a, b \in R$ .

☐

$ax = ay, \forall a \in R \implies x = y$ , where  $x, y \in V$ .

☐

None of the above.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$x + y = z + y \implies x = z \forall x, y, z \in V$ .

$ax = bx, \forall x \in V \implies a = b$ , where  $a, b \in R$ .

$ax = ay, \forall a \in R \implies x = y$ , where  $x, y \in V$ .

1 point

Which of the following options is/are true for a non zero vector space  $V$ ?

☐

$av = 0, \forall v \in V \implies a = 0$ .

☐


$$av = 0, \forall a \in R \implies v = 0 \quad av = 0, \forall a \in R \implies v = 0.$$

☐

$$c \in R, c \in R \text{ and } v, w \in V, v, w \in V \text{ such that, } c(v - w) \neq cv - cw, c(v - w) = cv - cw.$$

☐

$$c((v + w) - (v - w) - 2w) = 0 \quad \forall c \in R, \text{ and } v, w \in V, c((v + w) - (v - w) - 2w) = 0 \quad \forall c \in R, \text{ and } v, w \in V$$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$$av = 0, \forall v \in V \implies a = 0 \quad av = 0, \forall v \in V \implies a = 0.$$

$$av = 0, \forall a \in R \implies v = 0 \quad av = 0, \forall a \in R \implies v = 0.$$

$$c((v + w) - (v - w) - 2w) = 0 \quad \forall c \in R, \text{ and } v, w \in V, c((v + w) - (v - w) - 2w) = 0 \quad \forall c \in R, \text{ and } v, w \in V$$

Let  $(3, 2, -3) \in R^3$   $(3, 2, -3) \in R^3$  be a vector from the vector space  $R^3$   $R^3$  with usual addition and scalar multiplication. If  $c(3, 2, -3) = (-6, -4, 6)$   $c(3, 2, -3) = (-6, -4, 6)$ , find the value of  $c$ .

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) -2

1 point

1 point

Which of the following options is/are true?

☐ The zero vector is a unique vector in a vector space.

☐ Inverse of a vector is a unique vector in a vector space.

☐

$0v \neq 0v = 0$ , where  $v$  is a vector from a vector space of  $R$ .

☐

$c0 = 0c0 = 0$ , where  $c \in R, c \in R$  and  $0$  is the zero vector from a vector space over  $R$ .

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

The zero vector is a unique vector in a vector space.

Inverse of a vector is a unique vector in a vector space.

$c0 = 0c0 = 0$ , where  $c \in R, c \in R$  and  $0$  is the zero vector from a vector space over  $R$ .

## Level 2

Let  $V$  be a plane parallel to the  $XY$ -plane. Any plane parallel to  $XY$ -plane is given by  $z = c$ . We define addition of two vectors  $v_1 = (x_1, y_1, c)$  and  $v_2 = (x_2, y_2, c)$  on  $V$  as follows: First project  $v_1$  and  $v_2$  on the  $XY$ -plane (we will get the vectors  $(x_1, y_1, 0)$  and  $(x_2, y_2, 0)$  by projection on the  $XY$ -plane) and then calculate the addition of the vectors we obtained by the projection on the  $XY$ -plane (we will obtain  $(x_1 + x_2, y_1 + y_2, 0)$ ). Then project the obtained vector back to the plane  $V$  (we will obtain the vector  $(x_1 + x_2, y_1 + y_2, c)$ ).

Let  $V$  be a plane which is parallel to the  $XY$ -plane, defined by  $z = 2$ . Let  $v_1 = (1, 2, 2)$

$= (1, 2, 2)$  and  $v_2 = (0, 3, 2)$  be in  $V$ . Answer questions 7 and 8.

If  $v'_1 = (a_1, b_1, c_1)$  and  $v'_2 = (a_2, b_2, c_2)$  are the projections of  $v_1$  and  $v_2$  on the  $XY$ -plane, respectively, then find the value of  $2(a_1 + b_1 + c_1) - (a_2 + b_2 + c_2)$ .

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 3

1 point

If  $v_1 + v_2 = (a_1, b_1, c_1)$  as per the addition defined above, then find the value of  $2(a_1 + b_1 - 2c_1)$ .

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 3

1 point

Let  $V$  be a plane parallel to the  $XY$ -plane. Any plane parallel to the  $XY$ -plane is given by  $z = c$ . We define scalar multiplication of a vector  $v = (x, y, c)$  on  $V$  as follows: First project  $v$  on the  $XY$ -plane first (we will get the vector  $(x, y, 0)$  by projection on the  $XY$ -plane) and then calculate the scalar  $(\alpha)$  multiple of the vector we obtained by the projection on the  $XY$ -plane (we will obtain  $(\alpha x, \alpha y, 0)$ ). Then project the obtained vector back to the plane  $V$  (we will obtain the vector  $(\alpha x, \alpha y, c)$ ).

Let  $V$  be a plane which is parallel to the  $XY$ -plane, defined by  $z = 2$ . Let  $v = (1, 2, 2)$  be in  $V$ . Answer questions 9 and 10.

If  $v' = (a_1, b_1, c_1)$  is the projection of  $v$  on the  $XY$ -plane, then find the value of  $(a_1 + 2b_1 + c_1)$ .

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 5

1 point

If  $4v = (a_1, b_1, c_1)$  as per the scalar multiplication defined above, then find the value of  $(a_1 + 2b_1 + c_1)$ .

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 22

1 point

[Check Answers](#)

Your score is: 0/10

